DO NOW — RECALL

Floating-point arithmetic and precision

Name: _____ Class: _____ Date: _____

1. Normalise positive $0.0100_2 \times 2^4$.

2. What is −1 in 8-bit two's complement?

3. Calculate 0.5×2^4.

4. Calculate binary $0.1000_2 + 0.0100_2$.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

$0.1000_2 \times 2^3$.

2. Model answer

11111111.

3. Model answer

8.

4. Model answer

0.1100_2 = decimal 0.75.

Vocabulary — know these terms

exponent alignment — Express operands using the same power of two.

rounding — Choose a nearby representable value for an exact result.

truncation — Discard bits beyond the available precision.

overflow — Numerical result exceeds the representable range.

precision loss — Significant binary information cannot be stored.

NEW LEARNING

Read, annotate and explain

Align exponents

Before adding mantissas, both terms need the same power of two. Shift B right one place when raising its exponent by one.

Add the mantissas

$0.1000_2 + 0.0100_2 = 0.1100_2$; keep the common exponent 3.

Check in decimal

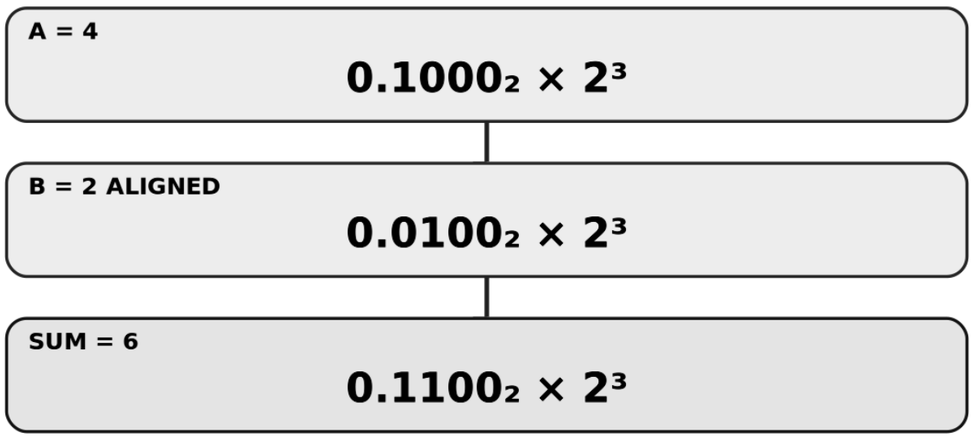
$0.1100_2 \times 2^3 = 0.75 \times 8 = 6$, matching 4 + 2.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read and interpret



A = 4, $0.1000_2 \times 2^3$; B = 2 ALIGNED, $0.0100_2 \times 2^3$; SUM = 6, $0.1100_2 \times 2^3$

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Add and check

0.1010 + 0.0010 = 0.1100 × 2³ = decimal 6.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

When increasing an exponent by 1 during alignment, the mantissa must shift right 1 place to preserve its value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

Calculate $0.1100_2 \times 2^3$ minus $0.1000_2 \times 2^2$.

Align

B=0.1000×2^2 becomes 0.0100×2^3.

Subtract

$0.1100 - 0.0100 = 0.1000$.

Check

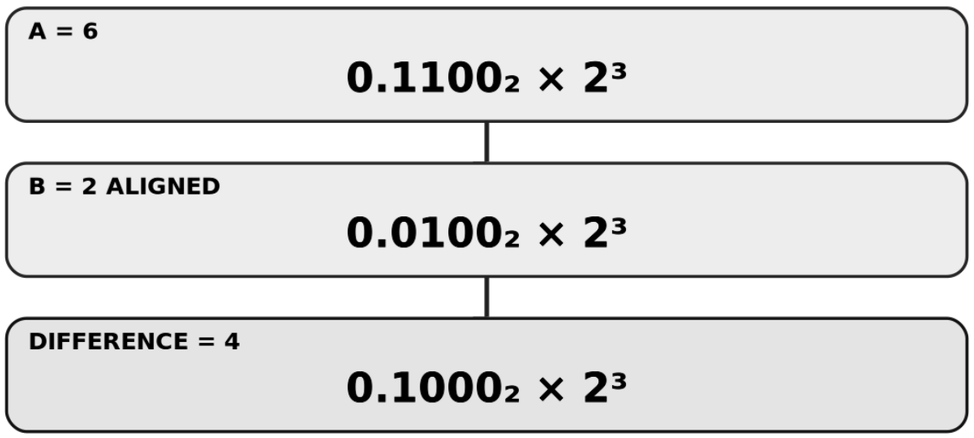
$0.1000_2 \times 2^3 = 4$; this matches decimal $6 - 2$.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Calculate $0.1100_2 \times 2^3$ minus $0.1000_2 \times 2^2$.

B=0.1000 × 2² becomes 0.0100 × 2³.



A = 6, $0.1100_2 \times 2^3$; B = 2 ALIGNED, $0.0100_2 \times 2^3$; DIFFERENCE = 4, $0.1000_2 \times 2^3$

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how exponent alignment can lose precision when the mantissa has a fixed number of bits.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Why must exponents be aligned before binary floating-point addition?

Question 2

Calculate and normalise $0.1010 \times 2^3 - 0.1000 \times 2^2$. Retain all shown mantissa bits during working.

WE DO — GUIDED PRACTICE

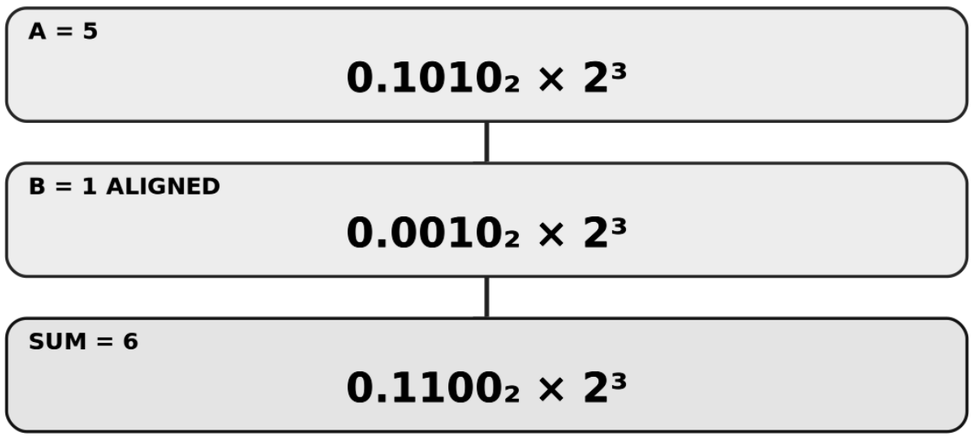
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Add $0.1010_2 \times 2^3$ and $0.1000_2 \times 2^1$.

Raise B's exponent 1→3 by shifting its mantissa right twice: 0.1000 becomes 0.0010.



A = 5, $0.1010_2 \times 2^3$; B = 1 ALIGNED, $0.0010_2 \times 2^3$; SUM = 6, $0.1100_2 \times 2^3$

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Align Raise B's exponent 1→3 by shifting its mantissa right twice: 0.1000 becomes 0.0010.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.